

Magnetising Magnesium

Chemistry Club 6th June

Magnetism

Which elements are magnetic?

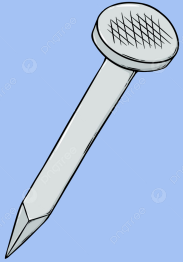
Cobalt (Co), Nickel (Ni) and Iron (Fe), in other words **not** magnesium?

What is a magnetic material?

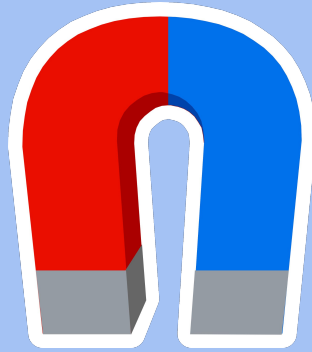
Materials which experience a force close to a magnet.



Magnetism



The iron nail is far away – it will not be attracted to the magnet – it will **not** experience a force.



Magnetism

The iron nail is in the bubble and **is** attracted to the magnet.

It experiences this attraction anywhere in the bubble.

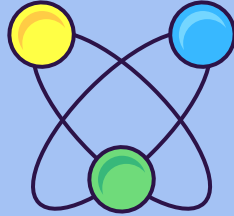


This bubble is the **magnetic field**.

Magnetic materials are materials which experience a force in the bubble.

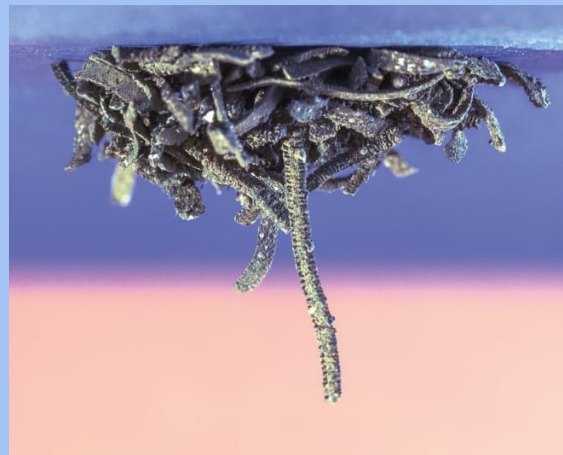


Method

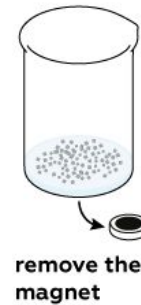
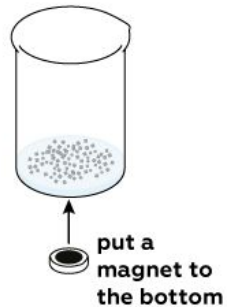
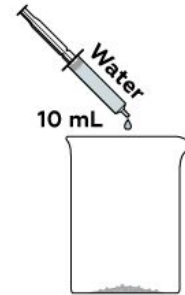
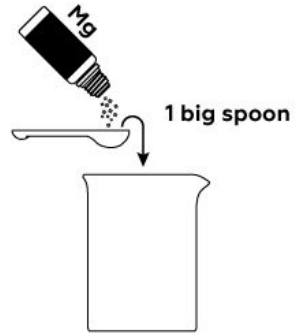


Method

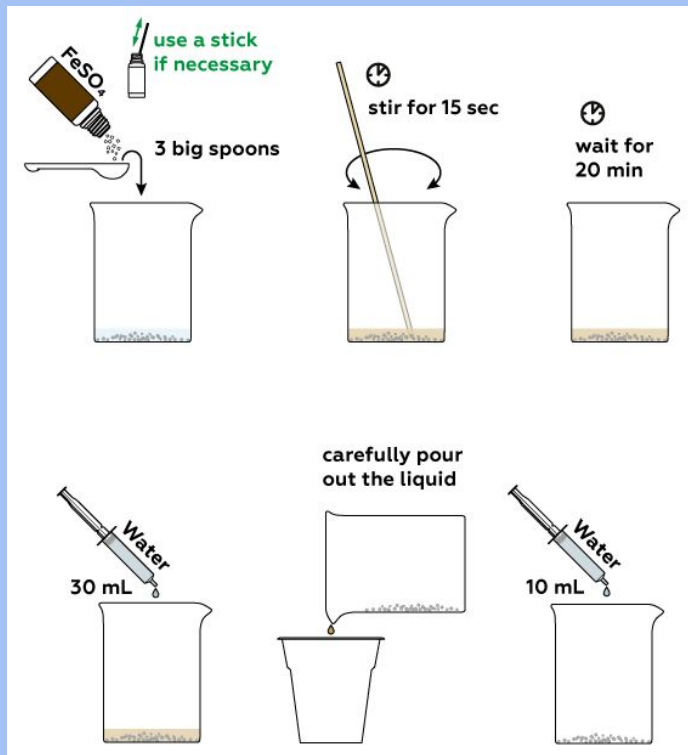
1. Add 1 spatula of Mg to the beaker.
2. Add 10 mL of water (5 pipettes).
 - a. If a magnet is placed beneath the beaker it **will not respond** .
3. Add 3 spatulas of Iron Sulfate.
4. Stir for 15 seconds. Allow to set for a few minutes.
5. Add 30 mL of water.
6. Then careful pour away the liquid (not the solid).
7. Now test with a magnet.



Method



Method



Method



the particles
respond to
the magnet



Why does it respond now?

Clues:

- Iron is a magnetic material.
- A reaction occurs, a reaction we've seen before ;)
- Our reaction mixture:



- Magnesium is **more** reactive than iron.

Answer:

Magnesium and Iron Sulfate react in a **displacement** reaction.

Fe and MgSO_4 form. Fe coats the surface and because Fe is magnetic it attracts.





So really, the magnesium is not magnetic :(

